DUALDIFF: ENHANCING MODE CAPTURE IN LOW-DIMENSIONAL DIFFUSION MODELS VIA DUAL-EXPERT DENOISING

Anonymous authors
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ABSTRACT

Diffusion models have demonstrated remarkable success in generating high-dimensional data, but their performance on low-dimensional datasets remains challenging, particularly in accurately capturing multiple modes. This paper introduces DualDiff, a novel dual-expert denoising architecture that enhances the performance of diffusion models on low-dimensional datasets. Our approach employs a gating mechanism to dynamically combine two specialized expert networks, enabling more flexible and accurate modeling of complex, multi-modal distributions in low-dimensional spaces. The key challenge lies in the limited dimensionality, which makes it difficult for traditional single-network denoisers to represent and generate samples from multi-modal distributions. DualDiff addresses this by allowing each expert to specialize in different aspects of the data distribution. We conduct extensive experiments on various 2D datasets, including 'circle', 'dino', 'line', and 'moons', demonstrating significant improvements in mode capture and sample diversity. Our method achieves a 38.7% reduction in KL divergence on the complex 'dino' dataset, from 1.060 to 0.650. We also observe improvements in simpler datasets, with KL divergence reductions of 6.2% for 'circle' and 3.1% for 'moons'. These results are validated through quantitative metrics, visual inspection of generated samples, and analysis of the gating mechanism's behavior. Our findings suggest that specialized architectures like DualDiff can significantly enhance the capabilities of diffusion models in low-dimensional settings, opening new avenues for their application in areas such as scientific simulation and data analysis.

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1 INTRODUCTION

Diffusion models have emerged as a powerful class of generative models, achieving remarkable success in generating high-dimensional data such as images and audio Ho et al. (2020); Yang et al. (2023). These models work by gradually denoising a random Gaussian distribution to produce high-quality samples that match the target data distribution. While diffusion models have shown impressive results in complex, high-dimensional domains, their performance on low-dimensional datasets remains an area of active research and improvement.

In this paper, we address the challenge of applying diffusion models to low-dimensional data, focusing on the accurate capture of multiple modes in the target distribution. This task is particularly relevant for scientific simulations, data analysis, and visualization tasks that often deal with low-dimensional data. Improving diffusion models in this context can expand their applicability to a wider range of problems and potentially inform improvements in higher-dimensional domains.

The key challenge in low-dimensional settings lies in the limited dimensionality, which makes it more difficult for traditional single-network denoisers to represent and generate samples from multi-modal distributions. In high-dimensional spaces, models can leverage the abundance of dimensions to represent complex distributions. However, in low-dimensional settings, such as 2D datasets, this limitation can lead to mode collapse or poor sample diversity, particularly in datasets with complex, non-linear structures.

To address this challenge, we propose DualDiff, a novel dual-expert denoising architecture for diffusion models in low-dimensional spaces. Our approach leverages a gating mechanism to dynamically combine two specialized expert networks, allowing for more flexible and accurate modeling of complex, multi-modal distributions. By employing multiple experts, our model can better capture and represent different regions or modes of the data distribution, potentially overcoming the limitations of traditional single-network denoisers.

The main contributions of this paper are as follows:

- We introduce DualDiff, a novel dual-expert denoising architecture for diffusion models, specifically designed to improve mode capture in low-dimensional spaces.
- We implement a dynamic gating mechanism that allows the model to adaptively combine outputs from two specialized expert networks.
- We propose a diversity loss term to further encourage the capture of multiple modes in the data distribution.
- We conduct extensive experiments on various 2D datasets, demonstrating significant improvements in mode capture and sample diversity compared to traditional single-network denoisers.
- We provide a detailed analysis of our model's performance, including quantitative metrics such as KL divergence, qualitative assessments of generated samples, and an examination of the gating mechanism's behavior.

Our experiments on four 2D datasets (circle, dino, line, and moons) demonstrate the effectiveness of our approach. Notably, our method achieves a 38.7% reduction in KL divergence on the complex 'dino' dataset, from 1.060 to 0.650. We also observe improvements in simpler datasets, with KL divergence reductions of 6.2% for 'circle' and 3.1% for 'moons' datasets. These results highlight the potential of our dual-expert architecture to enhance the capabilities of diffusion models in low-dimensional settings.

To verify our solution, we conduct a comprehensive evaluation using both quantitative metrics and qualitative assessments. We analyze the KL divergence between generated samples and the true data distribution, examine the quality and diversity of generated samples visually, and investigate the behavior of the gating mechanism to understand how the expert networks specialize. Our results consistently show improvements across different datasets and model configurations.

Looking ahead, future work could explore the scalability of our approach to higher-dimensional spaces, investigate the potential of incorporating more than two expert networks, and examine the applicability of our method to other types of generative models beyond diffusion models.

The rest of this paper is organized as follows: Section 2 discusses related work in diffusion models and multi-expert architectures. Section 4 details our proposed DualDiff architecture. Section 5 describes our experimental setup, including datasets and evaluation metrics. Section 6 presents and analyzes our results. Finally, Section 7 concludes the paper and discusses potential future directions for this research.

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2 Related Work

Our work on improving diffusion models for low-dimensional data builds upon several key areas of research in generative modeling and specialized architectures. Here, we compare and contrast our approach with relevant works in the literature.

2.1 Diffusion Models for Low-Dimensional Data

While diffusion models have shown remarkable success in high-dimensional domains Ho et al. (2020); Yang et al. (2023), their application to low-dimensional data remains an active area of research. The work of Kotelnikov et al. (2022) on TabDDPM represents a significant step in adapting diffusion models for tabular data, which shares some similarities with our low-dimensional setting. However, their approach focuses on handling mixed data types and high-dimensional tabular data,

whereas our method specifically addresses the challenges of capturing multi-modal distributions in low-dimensional spaces.

Karras et al. (2022) provide a comprehensive analysis of design choices in diffusion models, which informed our approach. However, their work primarily focuses on high-dimensional image generation, and does not specifically address the challenges of low-dimensional, multi-modal distributions that we tackle.

2.2 Multi-Expert Approaches in Generative Models

Our dual-expert architecture draws inspiration from mixture of experts models Goodfellow et al. (2016), adapting this concept to the diffusion model framework. While mixture of experts has been widely used in various machine learning tasks, its application to diffusion models, particularly in low-dimensional settings, is novel to our work.

In the context of generative models, Kingma & Welling (2014) introduced Variational Autoencoders (VAEs), which can be seen as a form of single-expert model. Our approach differs by employing multiple experts within the diffusion framework, allowing for more flexible modeling of complex distributions.

Similarly, Generative Adversarial Networks (GANs) Goodfellow et al. (2014) use a single generator network. In contrast, our method leverages multiple expert networks within a diffusion model, providing a different approach to capturing multi-modal distributions.

2.3 Techniques for Improving Mode Capture

The challenge of mode capture in generative models has been addressed through various techniques. Sohl-Dickstein et al. (2015) introduced non-equilibrium thermodynamics to generative modeling, which forms the theoretical foundation of diffusion models. Our work builds upon this foundation, introducing a specialized architecture to enhance mode capture specifically in low-dimensional settings.

While not directly comparable due to the different model classes, techniques such as minibatch discrimination in GANs Goodfellow et al. (2014) aim to improve mode capture. Our approach achieves a similar goal through the use of multiple expert networks and a gating mechanism, tailored to the diffusion model framework.

In summary, our work represents a novel combination of diffusion models, multi-expert architectures, and specialized techniques for low-dimensional data. Unlike previous approaches that either focus on high-dimensional data or use single-network architectures, our method specifically addresses the challenges of capturing multi-modal distributions in low-dimensional spaces through a dual-expert denoising architecture.

3 Background

Diffusion models have emerged as a powerful class of generative models, achieving remarkable success in various domains such as image and audio generation Ho et al. (2020); Yang et al. (2023). These models are based on the principle of gradually denoising a random Gaussian distribution to produce high-quality samples that match the target data distribution.

Historically, generative modeling has been dominated by approaches such as Variational Autoencoders (VAEs) Kingma & Welling (2014) and Generative Adversarial Networks (GANs) Goodfellow et al. (2014). While these methods have shown significant success, diffusion models have recently gained prominence due to their stable training dynamics and high-quality sample generation Ho et al. (2020).

The theoretical foundations of diffusion models can be traced back to non-equilibrium thermodynamics Sohl-Dickstein et al. (2015). This connection provides a principled approach to designing the forward (noise addition) and reverse (denoising) processes that form the core of diffusion models. Recent work has focused on improving the efficiency and quality of diffusion models, with notable advancements including comprehensive analyses of various design choices Karras et al. (2022).

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